THE CANONICAL CONORMAL-TO-DIFFERENTIALS SEQUENCE OF A FORMALLY UNRAMIFIED MAP OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3365-3376 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3378-3389, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). Feeds same-wave L16220.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas differentials-relative-immersion and differentials-universally-unramified; universal first order thickening machinery (in-source).

Source reference. Original lemma 05ZA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1145–1155. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04FB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3337–3347. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $h : Z \to X$ be a formally unramified morphism of algebraic spaces over B. There is a canonical exact sequence*

$$\mathcal{C}_{Z/X} \to h^*\Omega_{X/B} \to \Omega_{Z/B} \to 0.$$

The first arrow is induced by $d_{Z'/B}$ where Z' is the universal first order neighbourhood of Z over X.

Proof. We know that there is a canonical exact sequence

$$\mathcal{C}_{Z/Z'} \to \Omega_{Z'/S} \otimes \mathcal{O}_Z \to \Omega_{Z/S} \to 0.$$

see Lemma 05ZA. Hence the result follows on applying Lemma 04FB. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.